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TRUTH VALIDATED

Advanced EyeDetect Training

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VP

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Relevant-Comparison Test

Definition

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Relevant-Comparison Test (RCT)

- ▶ Primary or relevant issue (R1)
- ▶ Secondary or comparison issue (R2) – similar to probable truth
- ▶ Components of polygraph

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Why RCT?

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- ▶ Within-subject comparison
- ▶ R2 establishes a baseline
- ▶ Compare examinee reactions : R1 vs. R2



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Direct Interrogation

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- ▶ Test and process are standardized
- ▶ Test provides pre-test instructions / no interview
- ▶ Practice test
- ▶ Test



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Direct Interrogation

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- ▶ Test proctor only monitors the examinee, provides no instructions
- ▶ If there is test failure = examinee is guilty



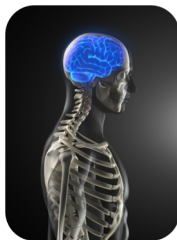
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RI Issue

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- ▶ Single Issue – one target behavior
- ▶ Multiple issue – up to 3 target behaviors combined into one issue



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Multi-issue Test

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- ▶ **Stealing**: Money? Products? Equipment? Min. value?
- ▶ **Drug use**: Which drugs? What time frame?
- ▶ **Serious crimes**: Examples? Definitions? As an adult?
- ▶ **Criminal ties**: cartels, gangs, organized crime, delinquents
 - ▶ Support, work with, and receive benefits from
- ▶ **Divulging confidential info** to unauthorized person
- ▶ **Bribes**: accepting or asking for? Money? Gifts? Favors?



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R2 issue requirements

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- ▶ Crime more serious than R1
- ▶ No cross over with R1
- ▶ Must have face validity for the examinee
- ▶ Test failure rate of 1-3%
- ▶ Should invoke a minimal response

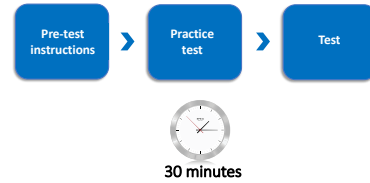


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Testing Process - direct

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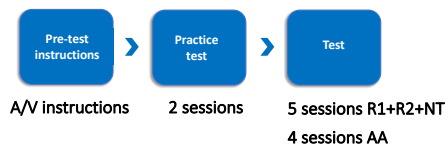


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Testing Process

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Indirect Interrogation

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Indirect Interrogation

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- ▶ Applicants are scarce (ex: US police)
- ▶ Rules might be bent
- ▶ Applicant asked to be 100% honest on the forms
- ▶ Are you **withholding information**?



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Indirect Interrogation

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- ▶ Withholding information test
- ▶ Relies on process and forms
- ▶ Pre-test interview needed



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Indirect Interrogation

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- ▶ Examinee must understand the difference between **being guilty** vs. **withholding information**
- ▶ Test failure = examinee guilty of withholding information



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Optimizing Results

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Standardize the process

- ▶ Gather information (forms)
- ▶ Video-based pre-test instructions



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Directed Lie Comparison

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Directed Lie

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- ▶ A transgression that most everyone will readily admit.
- ▶ During the test, the examinee is **directed** to lie to this question.
- ▶ We call them “**required to lie**” questions.

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Relevant Question

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- ▶ Question pertaining directly to the matter under investigation for which the examinee is being tested.



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Types of Relevant Questions

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- ▶ Primary relevant questions
 - ▶ Did you do it?
- ▶ Secondary relevant questions
 - ▶ Guilty knowledge
 - ▶ Secondary Involvement



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Primary Relevant Questions

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- ▶ Address the primary issue
- ▶ Test direct involvement
- ▶ Action verbs describe the act
- ▶ One per test



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Neutral Questions

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- ▶ Neutral or irrelevant - unrelated to testing issue
- ▶ Simple math questions

$$1 + 2 = 3$$

$$3 + 3 = 5$$

$$4 + 2 = 7$$

$$2 + 2 = 4$$

$$1 + 4 = 3$$

$$5 + 1 = 6$$

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Relevant Question Guidelines

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- ▶ Proper target selection
- ▶ Proper relevant question construction

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Target Selection Procedures

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Determine target issues, select most serious

- ▶ What did the examinee do?
- ▶ Probable motive
- ▶ What connects the examinee most closely to the crime?



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Target Selection Procedures

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- ▶ Did you shoot that man?
- ▶ Did you steal that car?
- ▶ Did you distribute those illegal drugs?



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Target Selection Guidelines

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- ▶ Don't ask about intent
 - ▶ Did you *purposely* touch that woman's breasts?
 - ▶ Did you *deliberately* set fire to that house?
 - ▶ Did you *intend* to hurt that child?



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Target Selection Guidelines

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Consider case facts and the target issue connecting the suspect to the crime

- ▶ Are there multiple suspects?
- ▶ Was suspect involved in several offenses?



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Target Selection Guidelines

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- ▶ Do you have evidence?
- ▶ Did the suspect admit anything?
If so, the questions change.



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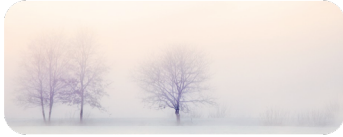
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Relevant Question

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I. Clear and concise

- ▶ Meaning must be unmistakable
- ▶ Consider examinee's IQ
- ▶ Less verbiage makes the question more clear



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Relevant Question

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Poorly constructed

- ▶ Did you steal money or anything else from the QuikMart yesterday?
- ▶ Did you touch any part of the female's sex organs for sexual gratification while at her house?



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Relevant Question

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If there are multiple items, use the phrase "any of"?

- ▶ Did you take any of the guns reported stolen?
- ▶ Did you take any of the money reported missing?



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Relevant Question

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2. Avoid emotion or judgmental terms

- ▶ Molest
- ▶ Mutilate
- ▶ Rip-off
- ▶ Butcher
- ▶ Kill

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Relevant Question

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3. Avoid legal terms

- ▶ Words that require two specific actions to occur to constitute the "legal definition"
 - ▶ Assault and battery
 - ▶ Aggravated Rape
 - ▶ Third-degree murder
 - ▶ Sodomize



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Relevant Question

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4. Present a dichotomy

- ▶ True or False
- ▶ The examinee's position must be clear.



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Relevant Question

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5. Should not be accusatory - avoid perception that the test has already reached a conclusion.

- ▶ After cashing the stolen check, did you spend any of the money?
- ▶ Did you use any of the cocaine you illegally bought?
- ▶ Did John tell you what he did with the gun after he shot the bank teller?

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Relevant Question

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6. Multiple items or amounts of money

- ▶ Money reported stolen
- ▶ If multiple items involved, use the phrase "any of."
- ▶ Reported amounts may be wrong.
 - ▶ Honest error.
 - ▶ Cover up of other crimes.



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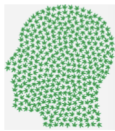
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Relevant Question

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7. May start or end with qualifiers

- ▶ Other than marijuana smoked that one time, did you ever...?
- ▶ Did you use cocaine between 1 June and 31 Sep. this year?



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Decision Model & Accuracy

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Algorithm

Logistic Regression Equation

$$\Pr(\text{Deceptive}) = 1 / (1 + \exp(b_0 + b_1X_1 + b_2X_2 + \dots + b_kX_k))$$

where X is an ocular-motor characteristic
where b is an optimal weight

Decision Rule

If $\Pr(\text{Deceptive}) < .50$, then "deceptive"
If $\Pr(\text{Deceptive}) > .50$, then "truthful"

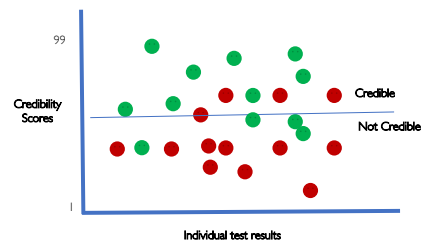


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Algorithm

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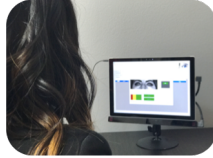
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Algorithm

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- ▶ Considers **16 features**: pupil dilation, blink rate, fixation, eye movement, error rate, response time, etc.
- ▶ Each variable (feature) is weighted
- ▶ Binary outcome – deceptive or credible



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Diagnostic Features

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- ▶ EyeDetect: pupil dilation



- ▶ Polygraph: EDA



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Validity and Reliability

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- ▶ Validity is accuracy.
- ▶ Reliability is repeatability over time.



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Validity and Reliability

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- ▶ Reliability impacts (constrains) accuracy.

- ▶ Both series have an average 2.5

- ▶ Series: 2, 2, 3, 2, 3, 3, 2, 3, 2, 2



- ▶ Series: 1, 4, 7, 3, 1, 1, 1, 1, 5, 1



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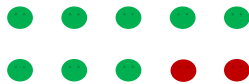
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Base Rate of Guilt

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Number of people in a testing group that are truly guilty. In the field, you will never know the BRG.

BRG = 20%



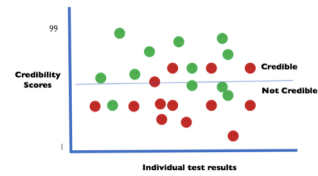
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Base Rate of Failure

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Number of people in a testing group that fail the test. It could include FP and TP.



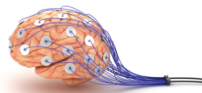
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Base Rates of Failure

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- ▶ In the lab, BRG is 50%
- ▶ Candidates for police officer: 33 - 43%
- ▶ Candidates for federal police: 60 - 70%
- ▶ Candidates for CBP: 90 - 95%

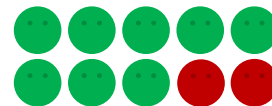


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BRG - 20%

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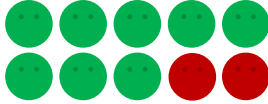
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BRG - 20%

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EyeDetect

- ▶ TP - 83%
- ▶ TN - 89%
- ▶ FN - 17%
- ▶ FP - 11%



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BRG - 20%

BRF - 30%

50

Results

- ▶ 7 innocent pass
- ▶ 1 innocent fails
- ▶ 2 guilty fail



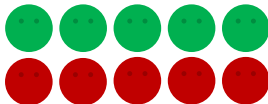
- ▶ All tests have errors

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BRG - 50%

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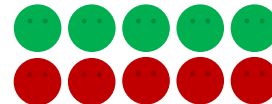
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BRG - 50%

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EyeDetect

- ▶ TP - 83%
- ▶ TN - 89%
- ▶ FN - 17%
- ▶ FP - 11%



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BRG - 50% BRF - 50%

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Results

- ▶ 4 innocent pass
- ▶ 1 innocent fails
- ▶ 1 guilty passes
- ▶ 4 guilty fail

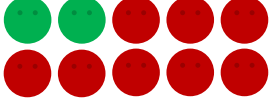


▶ All tests have errors

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BRG - 80%

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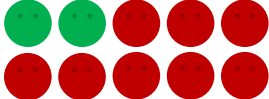
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BRG - 80%

55

EyeDetect

- ▶ TP - 83%
- ▶ TN - 89%
- ▶ FN - 17%
- ▶ FP - 11%



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BRG - 80% BRF - 80%

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Results

- ▶ 2 innocent pass
- ▶ 0 innocent fail
- ▶ 1 guilty passes
- ▶ 7 guilty fail



▶ All tests have errors

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Base Rates of Failure

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- ▶ Consider historical base rates for pass / fail
- ▶ Consider historical base rates for hire / don't hire



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BRF --> BRG

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- ▶ Base rates for pass/fail can help predict BRG
- ▶ Prior probabilities of guilt are important to decision making



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High Rates of Failure

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High Rates of Failure

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- ▶ Investigations can cause high rates of test failure
- ▶ Choose the right test question to avoid high failure rates.
- ▶ Poorly designed and broad questions have higher failure rates.

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High Rates of Failure

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- ▶ Large company suffers theft of \$10,000 in products from a storage facility, off-site in another location.
- ▶ Someone entered the storage facility's premises by using the company's password to enter a locked gate and then used bolt cutters to open the storage unit. The products were stolen at that point.



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High Rates of Failure

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- ▶ Company believes employees are involved because of the the passcode used to access the security gate.
- ▶ Employees tested
 - ▶ Had direct access to the storage facility
 - ▶ Had access to information about the storage facility
 - ▶ Worked with those with direct access
 - ▶ Managed the team that had access



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High Rates of Failure

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Test asked about 2 types of involvement:

1. Direct involvement in the theft or
2. Indirect involvement (helped someone else steal)



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High Rates of Failure

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High Rates of Failure

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An examinee can fail a test for having primary or secondary knowledge.

- ▶ **Primary knowledge:** direct involvement
- ▶ **Secondary knowledge:** indirect involvement in a variety of ways; also known as guilty knowledge



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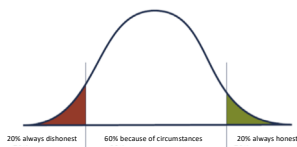
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Successive Hurdles

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Ethical behavior of employees

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- 60% behave dishonestly because of company conditions
- Primary reason: lack of internal controls and supervision

Source: KPMG 2015

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Multiple Hurdles

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- ▶ Various tests
- ▶ Applicants must pass each test

Step	Process
1	Application
2	Interview
3	Pre-placement examination
4	Physical fitness test
5	Medical evaluation
6	Psychiatric evaluation
7	Credibility assessment
8	Background investigation
9	Drug test (Urine)
10	Job offer made

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Successive Hurdles

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- ▶ Meehl and Rosen (1955)
 - ▶ Sensitivity - how well the test detects Guilt (TP)
 - ▶ Specificity - how well the test detects Innocence (TN)
 - ▶ Base rate - prevalence of condition such as P(Guilt) or P(Innocence)
- ▶ In diagnostic (psychometric) testing, all tests have error.
- ▶ Base rates of failure are vital to determining validity.
- ▶ Account for, manage, and take advantage of base rates
 - ▶ Improve Information Gain
 - ▶ Better understand test results

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Bayes Theorem Applied to Testing

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- ▶ Tests predict events.
- ▶ **Outcome Confidence** – a function of the test accuracies plus the base rates.
- ▶ **Base Rates** effect Outcome Confidence

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Bayes Theorem

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$$p(\text{deceptive} | \text{SR}) = \frac{p(\text{SR} | \text{deceptive}) * p(\text{deceptive})}{p(\text{SR} | \text{deceptive}) * p(\text{deceptive}) + p(\text{SR} | \text{truthful}) * p(\text{truthful})}$$

True Positives ÷ All Positives and account for the base rate of deception and truth telling.

$$p(\text{truthful} | \text{NSR}) = \frac{p(\text{NSR} | \text{truthful}) * p(\text{truthful})}{p(\text{NSR} | \text{truthful}) * p(\text{truthful}) + p(\text{NSR} | \text{deceptive}) * p(\text{deceptive})}$$

True Negatives ÷ All Negatives and account for the base rate of deception and truth telling.

Layman's definition – the probability of the 1st event affects the probability of the 2nd event.

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Base rate example

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- ▶ 500 Innocent and 500 Guilty = 50% BR
- ▶ Test is 90% accurate for both
- ▶ With equal BR, the confidence in the outcomes directly mirrors the accuracy of the test.

Equal Accuracy and Equal Base Rate of Guilt & Innocence

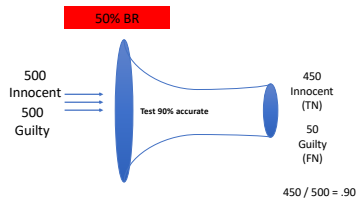
Ground Truth	Pass Test	Fail Test	Totals
Innocent	450 (TN)	50 (FP)	500
Guilty	50 (FN)	450 (TP)	500
Totals	500	500	1000
Outcome Confidence (NPV & PPV)	0.9 (NPV)	0.9 (PPV)	

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Bayes Theorem Applied to Testing

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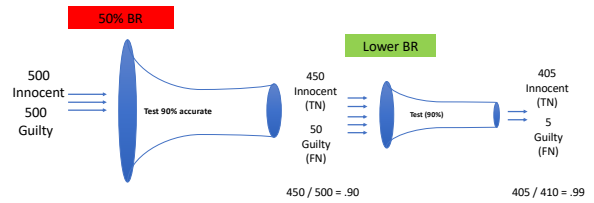


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Bayes Theorem Applied to Testing

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Accuracy – ODT and PDD

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	Pass Test	Fail Test
EyeDetect		
Innocent	0.89 (TN)	0.11 (FP)
Guilt	0.17 (FN)	0.83 (TP)
PDD		
Innocent	0.72 (TN)	0.14 (FP)
Guilt	0.08 (FN)	0.81 (TP)

Sources: Converus and APA Meta-Analytic Review (APA 2012, table 2)

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Polygraph peer-reviewed studies

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Table 4. Criterion accuracy profiles for evidentiary/diagnostic PDD techniques.

Technique	Federal You-Phase	IDCT*	MQTZCT*	Utah PLT (combined)	ZCT ESS
TDA Method	ESS	Horizontal	Matte	Utah	ESS
# Studies	2	3	3	7	6

Table 5. Criterion accuracy profiles for paired-testing techniques.

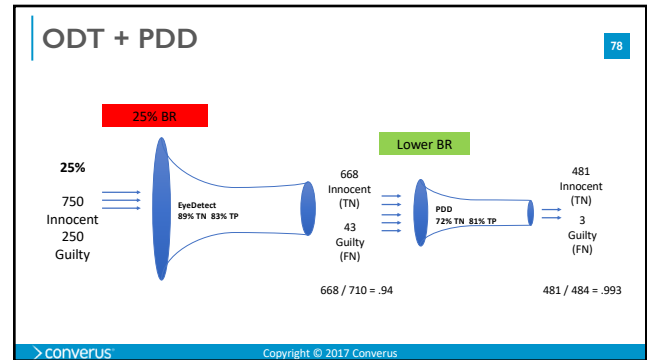
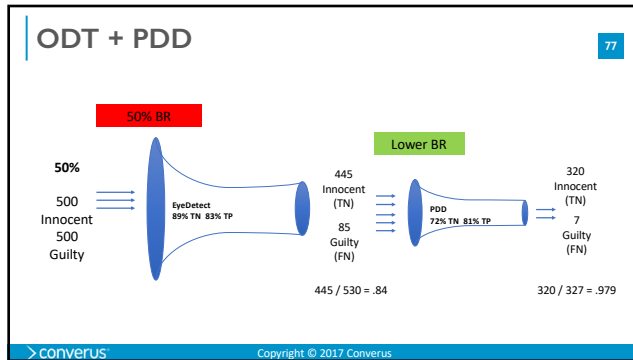
Technique	Backster You-Phase	Federal You-Phase	Federal ZCT	Federal ZCT	AFMGQT
TDA Method	Backster	7-position	7-position	7-position evidentiary	ESS
# Studies	2	2	3	2	3

Table 6. Criterion accuracy for investigative techniques.

Technique	CIT/GKT	DLST/TES	DLST/TES	AFMGQT
TDA Method	Lykken	7-position	ESS	7-position
Number of Studies	39	4	4	3

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Base Rate Considerations

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- ▶ Base rates affect confidence in outcome (NPV & PPV)
- ▶ There is no such thing as a perfect test.
- ▶ Make thoughtful estimates of base rates.
- ▶ Consider testing goals, budget, and base rates.
- ▶ Cost/benefit analysis

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Examples

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- ▶ Testing groups have different BRG
 - ▶ Sex offenders: 5 to 10%
 - ▶ Candidates for police officer: 33 - 43%
 - ▶ Candidates for federal police: 60 - 70%
 - ▶ Candidates for CBP: 90 - 95%

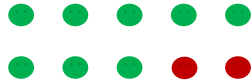
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Examples

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► Low BR group - more innocent fail (FP)



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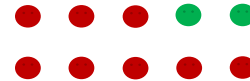
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Examples

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► High BR group - more guilty pass (FN)



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PDD and EyeDetect

"I want to compare the results of EyeDetect with polygraph."

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Study Requirements

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- ▶ Statistically valid sample size of examinees = 120 people
- ▶ Both test topics must be identical
- ▶ EyeDetect test precedes the polygraph test
- ▶ BRG must be estimated by the customer to set the algorithm properly
- ▶ Customer must report post-test admissions obtained by polygraph examiner
- ▶ Examinee confessions will be considered Ground Truth

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ODT + PDD

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- ▶ The probability the outcome of an EyeDetect test and a polygraph will be the same.

Examinee will...

- ▶ Pass both tests
- ▶ Fail both tests



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Standard deviation

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- ▶ 95% of all scores lie w/in 2 standard deviations of the mean.
- ▶ The mean accuracy of EyeDetect is .86 (point estimate).



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95% Confidence Interval

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- ▶ CI's are more informative than point estimates.
- ▶ The accuracy of a test varies within a range.
- ▶ Essentially it is a range of values that describes the uncertainty of the actual point estimate.



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DLST/ESS and EyeDetect

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DLST/ESS

TN is 75%
TN range 69 to 81%

TP is 81%
TP range 67 to 95%

Lab Study - EyeDetect

TN is 88%
TN range 82 to 94%

TP is 83%
TP range 81 to 85%

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Expected agreement

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DLST/ESS and EyeDetect

	TP	TN
Low	0.55	0.57
Medium	0.67	0.66
High	0.81	0.76

- TP should agree **55-81%** with mean est. of agreement of **67%**
- TN should agree **57-76%** with mean est. of agreement of **66%**

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AMFGQT/3-pos and EyeDetect

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AMFGQT/3-position

TN is **22%**
TN range **11 to 33%**

TP is 76%
TP range 70 to 82%

Lab Study - EyeDetect

TN is 88%
TN range 82 to 94%

TP is 83%
TP range 81 to 85%

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AMFGQT/3-position with 95% C.I.

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	TP	TN
Low	0.57	0.09
Medium	0.63	0.19
High	0.70	0.31

- TP should agree **57-70%** with mean est. of agreement of **63%**
- TN should agree **9-31%** with mean est. of agreement of **19%**

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Test Scores

Deceptive vs. Credible

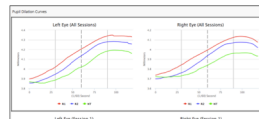
- ▶ 1 to 49 = Deceptive (DI) 
- ▶ 50 to 99 = Credible (NDI) 
- ▶ Example: 50 to 99 are "truthful" outcomes
 - 50 is like a +3 in PDD

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Deceptive behaviors during ODT

- ▶ **Pupil diameter:** greater among guilty
- ▶ **Response time on RI:** faster among guilty
- ▶ **Fixations:** fewer among guilty
- ▶ **First pass duration:** shorter among guilty
- ▶ **Blink rate:** lower for guilty
- ▶ **Error rate:** higher among guilty



Source: Lyn' Eyes Ocular-motor Measures of Reading Reveal Deception, Experiment 2, Journal of Experimental Psychology: Applied, Sep 2012

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Guidance Categories

✓ Credible	✗ Not Credible	⚠ Indeterminate
✓ Credible Truthful	✗ Not Credible Deceptive	⚠ Indeterminate Demo Test
	✗ Not Credible Confession	⚠ Indeterminate Aborted Test
	✗ Not Credible Probable Confession	⚠ Indeterminate No License
	✗ Not Credible Too Many Timeouts	⚠ Indeterminate Insufficient Data from Eye Scanner
	✗ Not Credible Random Responses or Low Comprehension	
	✗ Not Credible Countermeasure	

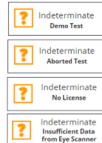
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Indeterminate

97

- ▶ **Demo test** – not scored; no test license
- ▶ **Aborted test** – test quit before completion
- ▶ **No License** – no test licenses
- ▶ **Insufficient data** – glare from glasses, dry eyes, excessive make-up or countermeasures



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Indeterminate

98

- ▶ **Insufficient data** – check room lighting



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Not Credible

99

- ▶ **Deceptive** – score of 1 to 49
- ▶ **Confession** – responding honestly to majority
- ▶ **Probable Confession** – responding honestly
- ▶ **Too Many Timeouts / Random Responses / Low comprehension** – poor reader or examinee with low IQ will miss questions and timeout randomly (no patterns)



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Countermeasures

100

EyeDetect Product Components

10
1

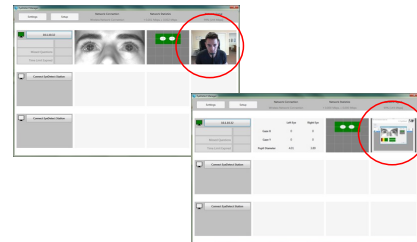
- ▶ **EyeDetect** – administer tests; installed on the EyeDetect Station
- ▶ **EyeDetect Manager** – observe examinees while testing; can be installed on any Windows device
- ▶ **Dashboard** – view test scores and reports, manage users and test licenses; cloud-based
- ▶ **EyeDetect Admin** – configure the tablet for use with subaccounts; installed on the EyeDetect Station
- ▶ **EyeDetect Video Recording** – video record examinees while testing; installed on the EyeDetect Station

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EyeDetect Manager

EyeDetect Manager

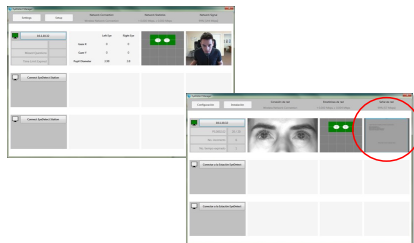
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EyeDetect Manager

EyeDetect Manager

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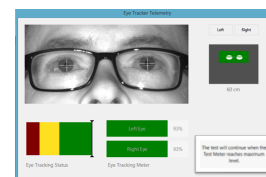
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Rest periods

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Re-calibration occurs automatically after each rest.



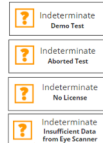
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Indeterminate

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5

- **Insufficient data**
 - Shutting eyes
 - Squinting eyes
 - Eye drops
 - Antihistamines (dry eyes)
- To-do: Monitor using EyeDetect Manager
- To-do: Check Telemetry screen at each break



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Countermeasures & Other Items

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- A confession generates a test result of "Not Credible"
- Algorithm for "Probable Confession" based on the Subject's responses
- Algorithm for "Random Responses" (which could happen with low comprehension, but is sometimes used as a countermeasure)
- Algorithm for "Too Many Timeouts" (which can be due to low comprehension, but is sometimes used as a countermeasure)
- Algorithm for real-time monitoring of "normal" pupil responses
- Algorithm for detecting unusual patterns (such as answer "False" to most of the RI's)
- Two diagnostic tests
 - Pupil Light Reflex Diagnostic
 - Digit Span Sequence Diagnostic
- Reading and comprehension test
 - Subjects who fail the reading test proceed directly into the ODT
 - If the Subject responds to the ODT as a "viable reader", we still score the OD

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Probable Confession Report (1 of 7)

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7

Test Response Detail (does not necessarily indicate credibility or deception) (R1)					Legend: ● Correct ● Incorrect ● Timed Out				
Statement	Correct Answer	S1	S2	S3	S4	S5	Self-incriminating		
I have not been involved in activities that would disqualify me for this job.	True	●	●	●	●	●	100%		
I have not participated in acts that disqualify me for this job.	True	●	●	●	●	●	80%		
I am not guilty of participating in the disqualifying activities mentioned.	True	●	●	●	●	●	100%		
I have never been involved in any of the disqualifying activities.	True	●	●	●	●	●	100%		
I am innocent of all involvement in the disqualifying activities.	True	●	●	●	●	●	100%		
Participating in disqualifying activity is something I have not done.	True	●	●	●	●	●	80%		
I have always resisted behaviors that would disqualify me.	True	●	●	●	●	●	100%		
My past does not include participating in activities that disqualify me from this job.	True	●	●	●	●	●	80%		
I have participated in activities that make me unfit for this job.	False	●	●	●	●	●	60%		
I have behaved in ways that would disqualify me from consideration for this job.	False	●	●	●	●	●	100%		
I am guilty of behaviors that would disqualify me from this job.	False	●	●	●	●	●	80%		
I have participated in activities that would disqualify me from employment.	False	●	●	●	●	●	80%		
I have engaged in an activity prohibited by the Dallas Area County Sheriff's Office.	False	●	●	●	●	●	80%		
My past includes participating in an act that would disqualify me for the job.	False	●	●	●	●	●	80%		
Participation in disqualifying activities is something I have done.	False	●	●	●	●	●	80%		
Participating in a disqualifying activity is a part of my past.	False	●	●	●	●	●	100%		

Note: Responses are highly self-incriminating.

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Probable Confession Report (2 of 8)

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Test Response Detail (does not necessarily indicate credibility or deception) (R2)					Legend: ● Correct ● Incorrect ● Timed Out				
Statement	Correct Answer	S1	S2	S3	S4	S5	Self-incriminating		
I have not worked with a gang or with an organized crime group in the past.	True	●	●	●	●	●	0%		
In my past, I have not been involved with a gang or organized crime group.	True	●	●	●	●	●	75%		
I am not guilty of previous involvement with a gang or organized crime.	True	●	●	●	●	●	0%		
I have never worked with any gangs or organized crime groups.	True	●	●	●	●	●	0%		
My past is free from involvement with gangs and organized crime.	True	●	●	●	●	●	0%		
Working with a gang or organized crime is something I have not done.	True	●	●	●	●	●	0%		
Associating with gangs or organized crime has not been one of my activities.	True	●	●	●	●	●	0%		
My past does not include involvement with gangs or organized crime.	True	●	●	●	●	●	0%		
I am guilty of involvement with a gang or with an organized crime group.	False	●	●	●	●	●	0%		
I have been recruited to work for a gang or an organized crime group.	False	●	●	●	●	●	0%		
I am guilty of involvement with someone in a gang or organized crime.	False	●	●	●	●	●	0%		
I admit to being involved with a gang or with an organized crime group.	False	●	●	●	●	●	20%		
At times, I have worked for a gang or for an organized crime group.	False	●	●	●	●	●	0%		
My past activities include involvement with a gang or organized crime.	False	●	●	●	●	●	0%		
Associating with a gang or organized crime is something that I have not avoided.	False	●	●	●	●	●	20%		
Involvement with a gang or organized crime has been a part of my past.	False	●	●	●	●	●	0%		

Note: Yet the Subject had no problem with the R2 statements.

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Tested positive or false positive

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What is a False Positive (FP)?

- ▶ EyeDetect is 86% accurate. No test is perfect.
- ▶ FP rate of EyeDetect is 11-12%.



Next steps:

- ▶ Test with polygraph
- ▶ Advise the examinee and take no action.
- ▶ Conduct a post-test interview.

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Current employee

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8

There are many possible courses of action.



- ▶ Take no action but monitor the person's work and work habits (possibly surreptitiously)
- ▶ Advise the person/don't interview further.
- ▶ Advise the person/interview further.
- ▶ Schedule a polygraph for a successive hurdles approach to minimize FPs.

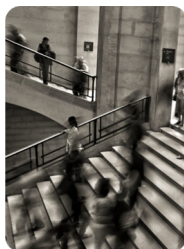
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Job applicants - plenty

11
9

- ▶ If the job applicant comes from a plentiful pool, perhaps it may be better to move that candidate further down the list for consideration to hire.
- ▶ We never advocate taking adverse action solely based on the test result, but initial resources may be better used on applicants who passed.



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Job applicants scarce

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0

- ▶ If time permits, conduct a more thorough investigation of that applicant.
- ▶ If time (and budget) permit, schedule a polygraph as a second hurdle to minimize FP errors and increase Outcome Confidence.



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Thank you